
Algorithm 4: Label-setting algorithm with UB pruning, scoring, and early stopping

In this initial version, we use the following scoring formula:

$$Score(L) = |C| + (1 - \tau)UB; \text{ where } \tau = \frac{t - \text{start_time}}{24:00}$$

Input:

- V . Set of nodes
- E_s . Set of scheduled edges (v, u, dep, arr)
- E_t . Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets
- $\text{bucket}(t)$. Time bucket mapping function
- record . Current record to beat (e.g. 13)

Output:

- P . Sequence of labels representing a route within 24 hours

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1: Initialize  $\text{Labels}(v) \leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize priority queue  $Q \leftarrow \emptyset$  // ordered by decreasing  $\text{score}, |C|, UB, -t$ 
3: Initialize  $\text{best\_label} \leftarrow \text{null}$ 
4: // Initialize labels at departures
5: for all  $v \in V$  do
6:   for all scheduled departure from  $v$  at time  $dep$  with arrival  $arr$  do
7:      $L \leftarrow (v, dep, \{\text{country}(v)\}, \text{pred} = \text{null}, \text{start\_time} = dep)$ 
8:      $\text{Labels}(v) \leftarrow \text{Labels}(v) \cup \{L\}$ 
9:     compute  $\text{score}(L)$ 
10:     $Q \leftarrow Q \cup \{L\}$ 
11: // Main loop
12: while  $Q \neq \emptyset$  do
13:   extract  $L = (v, t, C, \text{pred}, \text{start\_time})$  with maximum score from  $Q$ 
14:   // Enforce 24-hour journey limit
15:   if  $t - \text{start\_time} > 24 : 00$  then
16:     continue
17:   // Upper bound computation
18:    $b \leftarrow \text{bucket}(t)$ 
19:    $UB \leftarrow |R(v, b) \setminus C|$ 
20:   // Record-based pruning
21:   if  $|C| + UB \leq \text{record}$  then
22:     continue
23:   // Update best label
24:   if  $\text{best\_label} = \text{null} \vee |C| > |\text{best\_label}.C|$  then
25:      $\text{best\_label} \leftarrow L$ 
26:     if  $|C| > \text{record}$  then
27:       break // early stopping: record beaten
28:   // Extend via scheduled edges
29:   for all  $(v, u, dep, arr) \in E_s$  do
30:     if  $dep \geq t \wedge dep - \text{start\_time} \leq 24 : 00$  then
31:        $t_{\text{new}} \leftarrow arr$ 
32:       if  $t_{\text{new}} - \text{start\_time} > 24 : 00$  then
33:         continue

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34:    $C' \leftarrow C \cup \{\text{country}(u)\}$ 
35:    $L' \leftarrow (u, t_{\text{new}}, C', \text{pred} = L, \text{start\_time} = \text{start\_time})$ 

36:   if  $\nexists L'' \in \text{Labels}(u)$  such that
       $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C' \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C'))$  then
37:       remove all  $L'' \in \text{Labels}(u)$  such that
           $(t_{\text{new}} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C' \neq L''.C))$ 
38:        $\text{Labels}(u) \leftarrow \text{Labels}(u) \cup \{L'\}$ 
39:       compute score( $L'$ )
40:        $Q \leftarrow Q \cup \{L'\}$ 

41:   // Extend via transfers
42:   for all  $(v, w, \Delta) \in E_t$  do
43:        $t_{\text{new}} \leftarrow t + \Delta$ 
44:       if  $t_{\text{new}} - \text{start\_time} > 24 : 00$  then
45:           continue
46:        $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = \text{start\_time})$ 

47:       if  $\nexists L'' \in \text{Labels}(w)$  such that
           $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$  then
48:           remove all  $L'' \in \text{Labels}(w)$  such that
               $(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$ 
49:            $\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$ 
50:           compute score( $L'$ )
51:            $Q \leftarrow Q \cup \{L'\}$ 

52:   // Reconstruct path
53:    $P \leftarrow$  empty sequence
54:    $L \leftarrow \text{best\_label}$ 
55:   while  $L \neq \text{null}$  do
56:       prepend  $L$  to  $P$ 
57:        $L \leftarrow L.\text{pred}$ 
58:   return  $P$ 

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In Algorithm 4 we make a change that targets one of the main issues in Algorithm 2: the explosion of labels, which are rarely pruned due to the very relaxed conditions of dominance.

As seen in the notes, we now prioritize labels based on a scoring function that takes into account the number of countries visited, the Upper Bound of reachable countries from each node at a specific time (bucket) and the time of the label. This way we hope to expand more promising nodes first.

This is combined with an early stopping condition that breaks the loop as soon as we beat the record. Expanding promising labels earlier, combined with the early stopping, should result in a faster algorithm.

We also add a pruning condition such that labels where the sum of current number of countries and upper bound of reachable countries is less than the record are discarded. This also saves us time, since we don't care about solutions that are guaranteed to be worse than the record.